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Federated Learning in Healthcare: From Research to Real-World Deployment

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Keywords

federated learning, healthcare, big data, fairness, security, privacy

Abstract

Artificial intelligence (AI), including deep and traditional machine learning, holds great promise for advancing biomedical research and healthcare. However, most AI studies remain academic in nature and rarely transition into clinical practice, largely due to limited access to diverse real-world datasets. Centralized learning, the traditional approach to multi-institutional collaboration, is hindered by privacy, legal, and logistical barriers. Federated learning (FL) offers a decentralized alternative, enabling institutions to collaboratively train models without sharing sensitive patient data. This article reviews key algorithmic, privacy, and practical developments in FL for biomedical engineering, including strategies to handle non-identical data distributions and safeguard privacy through differential privacy, secure aggregation, and confidential computing. We also discuss current limitations and considerations for the need of scalable, interoperable infrastructures. FL represents a paradigm shift toward building generalizable, equitable, and clinically impactful AI models. Realizing this vision requires continued advances, such as FL-as-a-service platforms

and regulatory-aligned workflows that support persistent and trustworthy model deployment to truly realize AI’s promise in patient care.

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1. INTRODUCTION

The modern era of healthcare, medicine, and associated biomedical research is data-driven and complex. Health systems, from large academic medical centers to clinics in community settings, generate unprecedented amounts of heterogeneous data types from multiple sources, spanning across activity logs, clinical reports/records, radiology imaging, lab tests, pathology representations, and molecular profiling (1). The collection of these heterogeneous data represents the electronic health records (EHRs) of individual patients that need to be assessed collectively to allow optimal clinical decision-making. However, such assessment is cumbersome in the current clinical paradigm, where clinical experts manually navigate through the data, due to the multidimensional data generation, the required multidisciplinary expertise to evaluate these data, and the disconnect across siloed departments and data types.

Artificial intelligence (AI)—a term used to encapsulate both deep learning (DL) and traditional machine learning (ML)—has shown promise for healthcare applications across various diseases and anatomical sites, while leveraging unimodal and/or multimodal data types. However, such AI studies and methods have predominantly remained in the academic space (at best published as peer-reviewed manuscripts) rather than providing a direct impact on patient prospects and optimizing patient care, increasing the healthcare throughput, or improving clinical efficiency. The main limiting factor on moving these methods closer to their clinical translation has been the availability of large real-world datasets capturing patient populations of diverse demographic

AI: artificial intelligence

DL: deep learning

ML: machine learning

SUMMARY OF CENTRALIZED LEARNING CONCERNS

- Legal: Data are legally protected [e.g., the Health Insurance Portability and Accountability Act (HIPAA), General Data Protection Regulation (GDPR)].
- Technical: Data are too large to transmit.
- Ownership/cultural: Data are too valuable to share.
- Privacy: Data are too sensitive to share.
- Limited: Data size and diversity are limited, due to all these concerns.

characteristics. Such data could allow either the training of robust models or the validation of model generalizability (by virtue of their inherent heterogeneity) to optimally address clinical questions, while avoiding demographic bias, or even address rare diseases. Although it is widely accepted that AI model training is a “data-hungry” process, most studies report model training on well-curated data (what some might call “toy” data, e.g., acquired under ideal conditions, or acquired under a controlled environment, or acquired under strictly unified protocols of clinical trials), rather than representing real-world data settings capturing diverse acquisition equipment and protocols or even missing data across a few data types (2). Furthermore, most studies have focused on the technical validation of AI methods across various evidence levels—from expert opinions to retrospective data to randomized controlled trials—rather than targeting AI applications with potential to improve patient management and outcomes, or establishing diagnostic procedures suitable for clinical use and reimbursement, that is, generating higher-value, evidence-guided outcomes (3).

Establishing international multi-institutional collaborations is essential to move toward impactful and clinically relevant AI models, since large real-world datasets cannot be found within a single institution. It is widely accepted that (a) model utility/performance is affected by the sample size and (b) model generalizability is affected by the data diversity, whether this refers to equipment/protocol differences or population subgroups (e.g., racial, sexual) (4–7). The traditional and most prevalent paradigm for multi-institutional collaborations has been based on data sharing [i.e., centralized learning (CL)], where individual sites share their data to a centralized location following comprehensive data sharing agreements (8, 9). This CL paradigm is certainly better than single-institutional studies but is still cumbersome to scale globally and results in limited data due to various legal, privacy, technical, or even ownership/cultural concerns (see the sidebar titled Summary of Centralized Learning Concerns). To address these concerns that prevent effective data centralization, federated learning (FL) was introduced in healthcare in 2018 (10), as an alternative decentralized paradigm to facilitate multi-institutional collaborations without sharing patient data, while capturing diverse patient populations (11, 12). We illustrate the conceptual differences of CL and FL in **Figure 1**.

1.1. Motivation for Federated Learning

Below, we highlight key points that shape the utility/performance, generalizability, and scalability of biomedical AI models and that also motivated the emergence of FL as a transformative alternative to CL, set to overcome its limitations.

- Model utility/performance depends on the size of the training data.
- Model generalizability depends on the diversity of the training data.
- Training robust models applicable in general patient populations requires data beyond the numbers that a single institution can hold.

CL: centralized learning

FL: federated learning

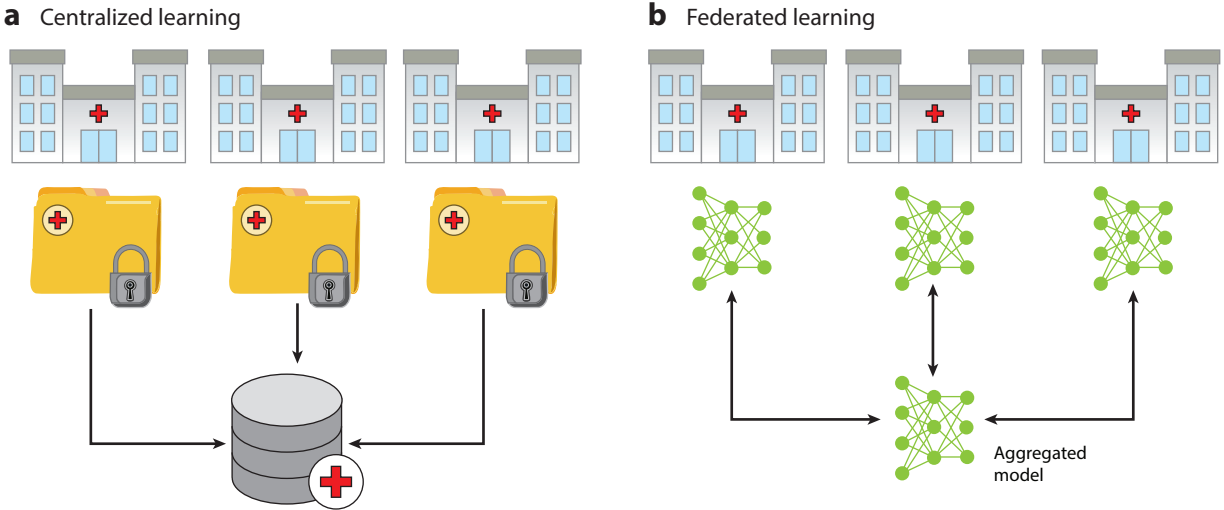


Figure 1

Comparison of (a) centralized learning and (b) federated learning. In centralized learning, data from multiple sources are aggregated in a central location for model training, raising privacy and data-sharing concerns. In contrast, federated learning enables collaborative model training across distributed data sources without transferring raw data, preserving privacy and enabling data sovereignty.

- Multi-institutional collaborations can address this.
- The current paradigm of CL does not scale due to various concerns, including data privacy and governance.
- FL represents a paradigm shift for multi-institutional collaborations addressing the CL-related concerns.

1.2. Article Scope and Outline

In this article, we (a) review, (b) present, (c) discuss, and (d) showcase algorithmic advances, security, privacy, and practical considerations of previous FL studies, as well as offer a future outlook on what is still missing to move FL to an operational stage for biomedical engineering applications. More specifically, Section 2 reviews and presents algorithmic advances focused on improvements furthering convergence, robustness, and communication of the federated averaging (FedAvg) method (13), particularly for configurations at collaborating sites where data follows non-identical and independent distributions (non-IID). Section 3 provides an overview of security and privacy considerations for FL. Section 4 focuses on future directions and what is currently missing to operationalize and scale FL in biomedical engineering. Finally, concluding summary remarks are presented in Section 5.

2. ALGORITHMIC ADVANCES

Standard AI algorithms require collecting data in a centralized database in advance. In FL, FedAvg (13) is a widely used method, which is based on decentralized gradient descent (14), and allows local nodes to perform more than one batch update on local data at a time and exchange updated model weights (e.g., gradients) for each update batch. To improve FedAvg on its convergence (15, 16), robustness (17, 18), and communication (19, 20), a number of FL studies have been conducted, particularly for clients with non-IID datasets. By examining mechanisms, applications,

Non-IID:
non-identical and
independent
distributions

and underlying principles of relevant approaches, this section aims to illuminate the current state of the art and provide a reference for researchers and practitioners seeking to deploy robust and effective FL systems in the complex landscape of healthcare.

2.1. Non-IID Issues in Biomedical Data Analysis

In the classical ML setting, the assumption of IID data is widely used to simplify model training. Due to the significant skew in data distribution across devices/locations, the IID assumption does not hold in FL anymore and presents a significant challenge for training performant models. There are typically two types of data skew: (a) skewed distribution of data labels and (b) skewed distribution of data appearance across devices/locations. Skewed data is a fundamental and pervasive problem for decentralized learning, causing significant accuracy loss in FL. The problem is especially difficult for DL models with batch normalization. The degree of data skew is a key determinant of the problem's difficulty.

2.1.1. Source of heterogeneity. A problem that occurs in many learning models in healthcare applications is the heterogeneity of the clients. The sources of the heterogeneity can be categorized as follows (and are illustrated in **Figure 2**):

- **Representation differences:** In the context of biomedical data, representation differences can come from the use of different medical imaging technologies, different preprocessing or feature extraction techniques, or different standards for data annotation. For instance, one hospital might use one type of magnetic resonance imaging (MRI) machine, while another hospital uses a different type. This can lead to differences in the appearance of the images, which can affect the learning process. Similarly, different healthcare systems might use different EHR systems that structure data in different ways. This can require extensive data harmonization efforts to ensure that the data can be used in an FL context.
- **Label distribution differences:** This refers to differences in the distribution of labels (outcomes) in the data from different locations. For instance, one hospital might serve a

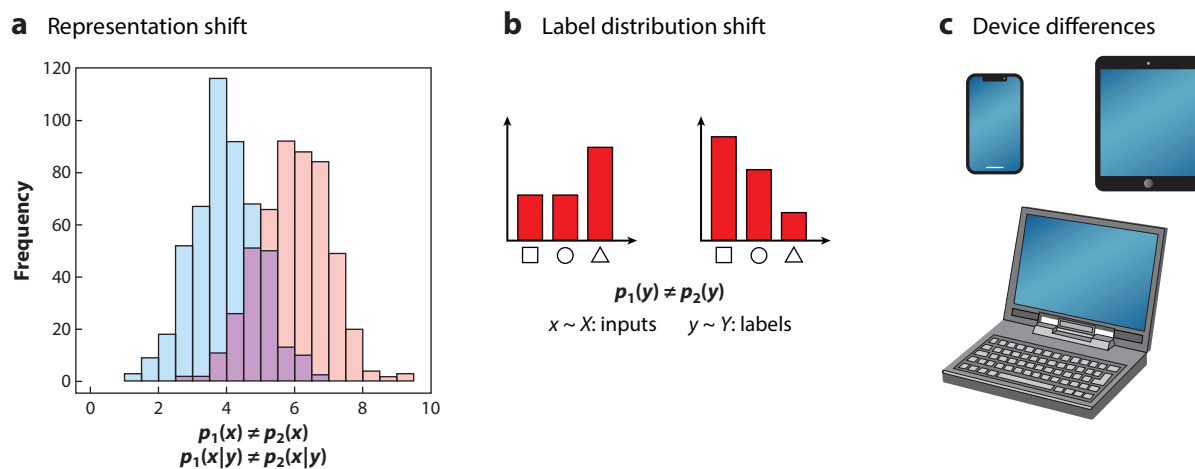


Figure 2

Illustration of non-identical and independent distributions challenges in federated learning, including (a) representation shift, (b) label distribution shift, and (c) device differences. The mathematical expressions highlight the differences in input and label distributions across clients.

population with a high prevalence of a certain disease, while another hospital serves a population where that disease is rare. Label heterogeneity can lead to a situation where the model learns to predict the disease well in the population where it is common but performs poorly in the population where it is rare. Strategies to handle this kind of heterogeneity can include the use of stratified sampling or reweighting methods to balance the label distribution during training.

- **Device differences:** In biomedical applications, the term devices could refer to the local servers or computing devices performing the local model training. These devices can vary in terms of their computational capabilities, the speed of their internet connections, and their availability for training (for instance, some devices might be available for training only during certain hours of the day). These differences can affect the speed and efficiency of the FL process. To handle this, FL algorithms can be designed to be robust to differences in device capabilities and availability. For instance, they can allow for asynchronous updates, where each device updates the model whenever it has completed a round of training, rather than requiring all devices to update the model at the same time.

2.1.2. Why and how heterogeneity affects federated learning. In FL, the training data are not required to be IID on the local devices. However, when the data are not IID, models trained in FL usually have worse performance than those trained in the standard CL mode. This is mainly because local models having the same initial parameters will converge to different models due to the heterogeneity in local data distributions. The divergence between the shared global model (acquired by averaging the uploaded local models) and the ideal model (the model obtained when the data on the local devices is IID) continues to increase, slowing down the convergence and worsening the learning performance (21).

The convergence of FedAvg (13) was first studied by Li et al. (22), revealing that the heterogeneity across clients slows down the convergence, which matches empirical observation. They also provide a necessary condition for FedAvg (13) to converge by a decaying learning rate with full gradient under convex and smoothness assumptions. Karimireddy et al. (15) identified a phenomenon they call client drift as the main reason for the poor performance of FedAvg in non-IID settings. Client drift refers to the divergence in the updates of each client, which results in slow and unstable convergence. This drift is particularly pronounced when the data are heterogeneous (non-IID). In a heterogeneous data scenario, the optimum of each client's loss function is likely to be far from the global optimum and from each other. As each client starts from the same point, their updates will move toward their respective client optimum. This means that the average of the client updates (which is the server update) moves toward the average of the client optima. To counter this drift, the commonly used FL method, FedAvg (13), is forced to use much smaller step sizes, which, in turn, hurts convergence. Xu et al. (23) identified the convergence issue being caused by catastrophic forgetting during the local training stage on each individual client, which refers to the severe forgetting of the knowledge of previous training data at the other clients, perhaps due to the discrepancy between local data distributions and the global data distribution.

As FL in biomedical engineering (and particularly imaging) applications typically falls in the cross-silo FL categories, we focus on non-IID data in this article. After understanding the challenges brought by non-IID data to FL, the next few sections of this article focus on the existing literature used to solve the problem of non-IID data in FL. Furthermore, we specifically focus on non-IID issues around representation differences and label distribution differences, since these are more prominently observed in biomedical engineering and imaging studies.

2.2. Local Client-Side Solutions

We begin by exploring client-side approaches to address data heterogeneity in FL, including data augmentation, regularized local optimization, meta-learning, and personalized FL methods.

2.2.1. Data augmentation. Data augmentation-based approaches aim to solve this problem by transforming or adjusting the data distributions in some way before the training. The most direct form of data augmentation involves applying a variety of transformations to existing images to artificially enlarge the training set and, more importantly, to balance skewed class distributions. Conventional techniques include geometric transformations such as rotation, flipping, and cropping, as well as intensity-based modifications such as altering contrast, adding Gaussian noise, or solarizing the image. To reduce data heterogeneity, some data-based approaches suggest sharing a global dataset among clients and combining global datasets with local datasets (24, 25). Duan et al. (26) used local augmentation to create a globally balanced distribution. FedMix extends the widely recognized mixup algorithm (27) from a centralized domain to an FL setting, which requires exchanging local data across clients for data interpolation on the input level, which raises privacy concerns. In contrast, FedFA (28) characterizes each client's data distribution as being able to be characterized by statistics of latent features and proposes to augment each local feature statistic probabilistically on the basis of a normal distribution whose mean is the original statistic and where the variance quantifies the augmentation scope. Therefore, it manipulates local statistics globally based on information in the entire federation to alleviate local data bias. Literature also reveals that leveraging generated data for data augmentation in FL is vulnerable to backdoor attacks (29), a common safety issue in FL (see Section 3).

2.2.2. Regularized local optimization. As analyzed in Section 2.1.2, the divergence among local models or between local and global models is the root issue for cumbersome convergence in FL. Applying regularization during local client training is a key technique to improve model performance with non-IID and unbalanced data (30).

The FedProx algorithm (17) stands as a foundational and widely adopted improvement over the standard FedAvg algorithm. Its approach to mitigating client drift is elegant in its simplicity by adding a proximal term to each client's local objective function, which takes the form of an L2 regularization penalty:

$$\frac{\mu}{2} \|w - w^t\|^2, \quad 1.$$

where w represents the local model weights being optimized, w^t are the weights of the global model received from the server at the start of the round, and μ is a crucial hyperparameter that dictates the strength of the regularization. The intuition is to penalize the local model for moving too far away from the global model.

SCAFFOLD emerged as another effective approach to employing gradient-correction-based regularization (15), preventing the clients' gradients from drifting away from the global gradient direction. Beyond the parameter-space constraints of FedProx and SCAFFOLD, other regularization strategies have emerged that target different aspects of the learning process.

One prominent example is FedBN (16), which focuses on regularizing the feature space to combat the feature shift common in multi-institutional medical imaging studies. For biomedical imaging, methods have been developed to improve both personalization and generalization. For instance, the GRACE method incorporates a feature alignment regularization on the client side, which corrects personalized gradients to prevent overfitting (31). Similarly, other frameworks use techniques such as batch adaptive normalization to allow client models to retain unique characteristics (32). LG-Mix (33) explicitly calibrates the local update direction by mixing with the global

gradient direction, showing promising results in medical image classification and segmentation tasks. Improving such client-side regularization methods is vital for the successful implementation of FL in real-world clinical scenarios (34, 35).

2.2.3. Meta-learning. Meta-learning is known as a “learning to learn” approach. Meta-learning has emerged as a powerful technique to handle data heterogeneity and improve personalization in FL for medical image analysis (36). A straightforward marriage between FL and meta-learning is FedMeta, where a parameterized algorithm or meta-learner is shared instead of the global model in most of the FL methods (37). In the biomedical domain, the proposed FedMetaMed framework, for example, has demonstrated superior performance compared with other state-of-the-art FL methods on real-world medical imaging datasets (38). Furthermore, frameworks such as RMPFD leverage meta-learning techniques within client clusters to optimize personalization, showing improved convergence and accuracy for biomedical imaging (39). Different from the strategies discussed earlier (augmentation, regularization, and distillation), which are all fundamentally aimed at producing and improving a single global model, meta-learning typically results in a federation of distinct client models. Therefore, the goal is no longer to force all clients, despite their heterogeneity, to agree on a single set of parameters. Instead, the goal is to leverage the collective knowledge of the federation to learn an optimal learning process (i.e., learn to learn), such as an adaptive learning rate schedule or a performance-based aggregation scheme, or to develop models that can quickly adapt to new tasks.

2.2.4. Personalization in federated learning. An important application of penalization in FL is using the model under different contexts (i.e., different tasks or distributions). As discussed by Kairouz et al. (40), the gap between the performance of global and personalized models shows the crucial importance of personalization in FL. The number of research studies in personalized FL has proliferated during the past few years. Several different approaches are trying to personalize the global model. This has significant implications for how FL could be deployed in clinical settings, moving away from the notion of a single, universally approved global model and toward a framework for creating “locally validated, globally informed” diagnostic tools. This better reflects clinical reality, where a model may need to be fine-tuned for a specific hospital’s unique environment, although it also introduces new complexities for regulation and validation. A very first personalized FL method is FedPer, first proposed by Reference 41, where each client contains a personal model and a base model. The FL server computes the average of all base models and then sends the averaged base model to each client. Then the data owner locally updates the base model together with their own personal model using their own private data. Later, adaptive personalized federated learning (APFL) (42) was advocated as an algorithm, where each client trains their local models while contributing to the global model. The dominant approach for personalization is local fine-tuning, where each client fine-tunes the received global model using its own local data by several gradient descent steps. This approach is predominantly used in meta-learning and transfer learning. However, in a different manner from these learning strategies designed for CL that try to learn multiple models, personalized FL mainly focuses on a single global model. Architecture-based approaches, such as the pioneering FedPer, achieve personalization by decoupling the model into shared and private components (43). This allows the model to learn global patterns through the federated part while capturing local data nuances in the private part. This strategy has been effectively applied in medical imaging, for instance, by personalizing only the decoder part of a model for medical image segmentation (44) and MR image reconstruction (45), allowing it to adapt to client-specific image details. Another example is FedDP, which personalizes the query embeddings in a transformer-based model for medical image segmentation, enabling clients to learn long-range relationships across the entire federation’s data (46).

2.3. Global Server-Side Solutions

While some solutions operate on the client side, a powerful and increasingly sophisticated class of methods involves interventions orchestrated by the central server. These server-side solutions leverage the server's unique global perspective to intelligently manage the FL process, steering it toward a more stable and effective convergence. This section provides an exhaustive survey and analysis of three principal categories of server-side strategies designed to combat the adverse effects of non-IID data in federated biomedical analysis: strategic client selection, client clustering for groupwise federation, and the use of proxy data and knowledge anchors.

2.3.1. Client selection. In a typical FL workflow, the central server randomly selects a subset of available clients to participate in each communication round. While simple and unbiased, this random sampling approach can be highly inefficient in the presence of non-IID data. The aggregation of updates from a randomly chosen, statistically heterogeneous group of clients can introduce significant variance into the global model, leading to the well-documented problems of client drift, slow convergence, and poor final accuracy (15). The core issue with random selection in non-IID settings is that the expected gradient over a random subset of clients may not align with the true global gradient over the entire population. This discrepancy, exacerbated by multiple local update steps, causes local models to diverge toward their respective optima, a phenomenon termed client drift. Strategic client selection emerges as a powerful server-side countermeasure. Instead of passive random sampling, the server actively curates the cohort of participating clients in each round based on specific criteria related to performance, similarity, or resource availability. This intelligent orchestration allows the server to steer the global learning process, manage the impact of client drift, and optimize for faster and more stable convergence. To reduce the interclient data heterogeneity within a single training round, similarity-driven client selection methods construct a statistically homogeneous training cohort for each round, for example, by measuring clients' similarities (46). Another class of methods uses the model updates or gradients as a proxy for the underlying data distribution. The server can calculate the similarity between the weight updates from different clients to identify those that are moving in similar directions in the parameter space. Building on this, the FEDACS algorithm employs a server-side attention mechanism (47). This allows the server to learn the relevance of clients to one another and selectively aggregate models based on these learned attention weights, effectively enhancing collaboration among clients with similar data distributions. Furthermore, client selection can be reframed as a form of implicit, server-side regularization. While client-side methods, such as FedProx, add an explicit regularization term to the local loss function to penalize divergence from the global model, client selection achieves a similar outcome at the aggregation stage.

2.3.2. Cluster learning. While strategic client selection refines the training process on a round-by-round basis, client clustering takes a more structural approach to handling data heterogeneity. Instead of assuming all clients will eventually converge to a single global model, clustered federated learning (CFL) acknowledges that in highly heterogeneous environments, such as multi-institutional medical networks, distinct subgroups of clients may exist. This is particularly true in medical imaging, where data from different hospitals can form natural clusters due to specializations (e.g., pediatric versus adult hospitals), equipment vendors, or patient populations. The core principle of CFL is to partition the client population into clusters based on the similarity of their data or model characteristics and then train specialized models for each cluster. This allows the resulting models to specialize to the particular data characteristics of their respective clusters, often leading to significantly improved performance compared with a one-size-fits-all global model (48, 49). There are several metrics for client clustering in FL. Model-update-based clustering uses the

behavior of the client models during training as a proxy for their underlying data similarity. The intuition is that clients with similar data distributions will produce similar model updates (i.e., gradients or weight differences) when training on the same global model (50). Other methods directly leverage the data embedding distances and then apply clustering algorithms (44). Across the literature, hierarchical clustering frequently emerges as the preferred algorithm for CFL. Its primary advantage in the federated context is that the number of clusters does not need to be predefined—a crucial feature when the underlying structure of the multi-institutional data is unknown (51). Overall, CFL occupies the space in between and offers “group-level personalization,” the granularity of which is directly controlled by the clustering algorithm and its parameters.

2.3.3. Proxy anchors. The most straightforward incarnation of this strategy involves the server holding a small, representative dataset. This proxy dataset, which could be publicly available or a small subset of IID data, can be used in several ways. Instead of exchanging concrete model parameters (weights and gradients), knowledge distillation (KD) approaches facilitate the transfer of knowledge in a more abstract form. In a federated context, this typically involves multiple client models acting as an ensemble of “teachers” that collaboratively train a central “student” model held by the server. This approach not only provides a mechanism to handle data heterogeneity but also can enhance communication efficiency and privacy. A significant practical challenge is the requirement for a suitable proxy dataset for the distillation process.

In many domains, especially medicine, a publicly available dataset that is representative of the clients’ sensitive, private data simply does not exist. This limitation has driven the development of “data-free” distillation methods that do not rely on an external dataset. A crucial advancement for privacy and practicality, especially in medical domains, is data-free knowledge distillation (DFKD). DFKD eliminates the need for the server to hold any real proxy dataset. Instead, the server learns to generate its own synthetic data to facilitate the distillation process, making the entire framework self-contained and more secure (52). FedFTG (53) is a data-free KD framework where the server trains an auxiliary generator to produce pseudodata. This synthetic data is then used as the medium to distill knowledge from the local teacher models into the global student model. To ensure the distillation is effective, FedFTG employs a hard sample mining scheme, adversarially training the generator to find inputs where the predictions of the local models and the global model disagree the most. Specifically for medical imaging, where feature distributions can be less discriminative than in natural images, using a pretrained generator becomes less effective. FedLGD (54) integrates dataset distillation algorithms into the FL pipeline to generate synthetic data for the medical domain and train FL using a smaller synthetic dataset. Specifically, to harmonize the domain shifts, FedLGD proposes iterative distribution matching to inpaint global information to local synthetic data and use federated gradient matching to distill global synthetic data that serve as anchor points to rectify heterogeneous local training without compromising data privacy. Similarly, the FedISCA (55) framework was proposed for one-shot FL. It uses an image synthesis technique (based on DeepInversion) on the server to generate a diverse set of images, ranging from random noise to realistic samples. The knowledge of the decentralized client models is then distilled into the global model using these synthetic data. Both FedLGD and FedISCA utilize models’ batch normalization statistics to the synthetic noise, a step that significantly improves KD performance, particularly in the early stages of synthesis when the generated biomedical images are less realistic.

3. SECURITY AND PRIVACY

We would first like to distinguish between the concepts of security and privacy. Security refers to protecting a system from unauthorized access or attack, while privacy refers to safeguarding personal information from being disclosed without consent.

Security and privacy are equally critical considerations in building scalable and secure distributed AI systems, such as those used for FL, which involves training AI models across multiple devices/nodes while keeping the data locally on these devices/nodes. The FL system must be secure enough to prevent unauthorized access or attacks while preserving data privacy during model training or even performing other types of analytics on distributed data. In FL, data are stored on devices or nodes rather than being transferred to a central server. This approach helps preserve privacy by keeping data on the devices, thereby maintaining the data governance of the data custodians. At the same time, the FL system should be secure enough to prevent unauthorized access to the data or the model.

Encryption and secure communication protocols must be used to achieve security and privacy in FL. Encryption can be used to protect data as it is transmitted between devices and the central server. Secure communication protocols can help prevent unauthorized access to data or models while in transit.

Therefore, security and privacy are closely intertwined and both critical considerations in building production-ready FL systems. By using techniques such as encryption (56), differential privacy (57–62), confidential computing via trusted execution environments (63–65), and secure communication protocols, both security and privacy can be achieved to protect the system from unauthorized access or attacks. Here, we provide a brief review of security and privacy considerations in FL. For a more in-depth analysis—including a taxonomy of threats and mitigation strategies and their performance implications—readers are encouraged to consult References 66–68 as complementary resources.

3.1. Security

Security in FL systems is multifaceted. Here, we outline the primary considerations that should be taken into account when building production-ready FL systems.

3.1.1. Information security and cybersecurity. Information security and cybersecurity are essential for protecting computer systems and data from unauthorized access, theft, or damage. For example, encryption can encode information to prevent unauthorized access. Firewalls are network security systems that control and monitor incoming and outgoing network traffic. Intrusion detection systems are designed to detect and respond to attempts to breach computer systems or networks. Access controls are used to restrict access to data and computer systems to authorized users only. In combination, these systems should be used to achieve effective information exchange and cybersecurity in FL.

3.1.2. Containerized environments. To improve the reliability of software applications, containerized environments such as Docker, Singularity, and MLCube can be used. These containers provide a reproducible method for packaging software applications and their dependencies in a portable and scalable format. Containers provide a virtualized environment that isolates the application from the host system, making it more secure and easier to manage. By packaging the application and its dependencies together, containers simplify the deployment process, ensuring consistency and reducing the risk of configuration errors. Additionally, container images can be versioned, providing a way to roll back to previous versions if issues arise, and can be helpful in production environments of FL systems.

3.1.3. Signed software code. Signing the code to be deployed on a client's system is a way to verify the authenticity and integrity of software applications. Signing code involves digitally signing the software with a unique identifier that users or other software systems can ascertain. This identifier ensures that the code has not been tampered with or modified in any way. By

signing the software code, users can have greater confidence in the authenticity and security of the application, reducing the risk of malware and other security threats.

3.1.4. An alternative categorization lens. Another angle to summarize the main security considerations in FL is provided below, following Reference 69:

- **Identity security:** Robust identity security requires both authentication and authorization mechanisms. Authentication ensures that entities are who they claim to be, using techniques such as passwords, biometrics, or public key infrastructure–based credentials. Once authenticated, role-based authorization policies should be enforced to control what each participant is allowed to do—for example, who can initiate experiments, deploy custom code, or shut down the system. In most healthcare FL use cases, only authorized and authenticated parties should be permitted to participate, ensuring compliance with regulatory, privacy, and institutional requirements (63, 70).
- **Communication security:** Using state-of-the-art end-to-end encrypted communication protocols such as transport layer security (71) is required to ensure the data's confidentiality.
- **Model protection:** In situations where trust is limited, the shared model information should be protected against attacks, such as gradient inversion methods that aim to reverse-engineer the training data from the shared model weights.
- **Data privacy:** Only the “knowledge” in the form of model weights or updates should be exchanged, not the underlying data. Of course, the exchanged models need protection considerations, as mentioned in the above bullet point. Additional privacy-enhancing techniques, such as differential privacy, should be applied if the models are prone to memorizing underlying data points from the training data (58).
- **Physical security:** This entails ensuring that physical access to FL systems and devices is limited to only essential personnel.
- **Firewall policies:** The access to network traffic (ports and internet protocol addresses) should be limited depending on the FL study.
- **Data protection policies:** All regional data management, storage, and access policies should be followed.

3.2. Attacks

FL allows multiple participants to collaboratively train a global consensus model without sharing their raw data. However, this collaborative approach can also introduce new security and privacy challenges. A comprehensive taxonomy of threats coupled with mitigation strategies is presented in Reference 66. In FL, different types of attacks can be launched to compromise participants' privacy or the model's accuracy. Privacy attacks focus on exploiting system vulnerabilities to infer sensitive information about participants or their data. Utility attacks, on the other hand, aim to manipulate the model's behavior or evade its detection capabilities. Understanding these types of attacks is crucial for building secure and robust FL systems. By implementing appropriate security and privacy measures, organizations can mitigate these risks and ensure the effectiveness of their FL consensus models.

In the following, we briefly define different types of privacy and utility attacks relevant to FL scenarios, following the categorization by Usynin et al. (72). By understanding these types of attacks, organizations can take steps to mitigate them and improve the security and utility of their FL systems.

3.2.1. Privacy attacks.

- **Model/gradient inversion:** An attacker tries to reconstruct the training data used to train the model or gradients by analyzing the model weights. This can be used to infer sensitive

information about the participants (62, 73–75). While theoretically possible, these attacks are often challenging to carry out in practical FL scenarios due to existing cybersecurity measures, high computational costs, and the strong assumptions about the attacker's prior knowledge (62).

- Membership inference: An attacker tries to determine whether a particular data point was used to train the model by analyzing the model output. This can reveal sensitive information about the participants and the data they contributed (76, 77).
- Attribute inference: An attacker tries to infer sensitive attributes of a participant from the model output, such as age, gender, or medical conditions (78, 79).
- Model inference: An attacker tries to recreate a replica of the original model by analyzing the model updates exchanged during the FL process (72).
- Side-channel attack: An attacker exploits weaknesses in the system to gain access to sensitive information, such as model parameters or training data (72, 80).

3.2.2. Utility attacks.

- Model poisoning: An attacker tries to inject malicious data into the training data to manipulate the model's behavior during the training process (81).
- Backdoor insertion: An attacker tries to add a back door to the model, which can be triggered by a specific input or pattern of inputs, allowing the attacker to control the model's output (82).
- Evasion attack: An attacker tries to manipulate the input data to evade the detection or classification of the model, which can lead to incorrect results and undermine the model's effectiveness (83, 84).

3.3. Privacy

One of the core tenets of FL is the safeguarding of private data. Despite its ability to obfuscate the necessity of consolidating datasets, the extent to which information exchanged during FL communication can be reverse-engineered remains uncertain (85). Therefore, it is essential to consider additional protection of the exchanged model information in FL. Techniques such as gradient encryption, obfuscation, and differential privacy can be effective in reducing the amount of patient-identifiable information encoded in the gradients exchanged during FL (62).

Obfuscating gradients is a technique that can be used to prevent gradient inversion attacks in FL. This technique involves adding noise to the gradients before they are shared between the participating entities, making it more difficult for attackers to reconstruct the original training data. There are several ways to obfuscate the gradients in FL. One approach is to use differential privacy, which involves adding controlled noise to the gradients to ensure that they are indistinguishable from the gradients obtained if a single participant's data is used for training or not. By adding noise to the gradients in this way, differential privacy makes it more difficult for attackers to reconstruct the original training data, thus protecting the participants' privacy. Differential privacy can be tuned to balance the participants' privacy with the model's accuracy (57).

Another approach is to use secure aggregation, which involves aggregating the model updates from the participants in a secure and private manner. Secure aggregation can be achieved using cryptographic protocols that ensure the model updates are combined without revealing sensitive information about the participants or their data (86).

Homomorphic encryption can also be used to protect the gradients in FL. Homomorphic encryption enables computations to be performed on encrypted data without decrypting it, which can be helpful for securely aggregating model updates without revealing sensitive information (56).

Another promising approach to enhancing privacy in FL is confidential computing, which leverages hardware-based trusted execution environments (TEEs) to protect data during computation. TEEs provide isolated regions of memory where code and data are securely processed and are shielded from the host operating system and other applications. In the context of FL, confidential computing can be used to perform gradient computations and model aggregation within a secure enclave, ensuring that sensitive information remains encrypted and inaccessible to unauthorized parties—even in the presence of a compromised system. This hardware-level protection adds a strong layer of defense against gradient leakage and other privacy attacks, complementing software-based techniques such as differential privacy and secure aggregation (87).

By obfuscating the gradients using one or more of these techniques, FL systems can be made more secure against privacy attacks. However, it is essential to utilize these techniques to strike a balance between privacy and model accuracy. For example, adding too much differentially private noise to the gradients can lead to degraded model performance (86, 88, 89).

4. CONSIDERATIONS FOR SCALING FEDERATED LEARNING

4.1. Technical Considerations

Implementing FL securely and on a scale has traditionally required deep technical expertise, custom infrastructure, and complex coordination between parties with expertise in different areas. Data custodians have a deep understanding of their data but might not have the expertise in participating in distributed or federated training experiments. Model developers often possess data science and ML expertise but may lack experience in managing a secure, fault-tolerant, distributed infrastructure for training their models. Neither model developers nor data custodians typically have the expertise in mitigating security and privacy risks introduced in an FL system, such as model intellectual property (IP) theft, data extraction attacks, model poisoning attacks, and protection from malicious data owners or malicious aggregators (see Section 3).

This complexity could be reduced by delivering a federated AI system as a service. This approach offers the benefits of AI training, validation, and analytics by facilitating cross-organizational collaboration, improving model quality, and ensuring data sovereignty, while alleviating the operational and technical burdens from participants.

4.1.1. Hosted versus on-premise infrastructure. In a hosted model, an FL service provider would manage most of the underlying infrastructure for FL, reducing the need for organizations to build and maintain their own federated stack. This includes secure result aggregation, secure communication, and workload orchestration (90). The service enables data providers to easily set up local (on-premises and public cloud) infrastructure for training, validation, and analytics, without requiring expertise in infrastructure or security. The service provider should implement zero-trust principles and be outside the trust boundary. Some highly sensitive use cases may warrant an on-premise deployment where there is greater control. However, this requires more expertise to configure and manage the FL system correctly.

4.1.2. Unified management interface. The federated system should have a web-based “single-pane-of-glass” portal that allows users to easily manage everything from dataset and experiment configuration to access control and monitoring. It must streamline workflows for data owners, model owners, and administrators, allowing them to easily collaborate and focus on building AI models rather than managing an FL system.

4.1.3. User and role management. Built-in identity and access management must allow organizations to define users with different roles, such as data owners, model owners, and administrators. This allows for clarity in responsibilities and helps enforce governance policies.

4.1.4. Confidentiality and integrity. The service should leverage privacy-enhancing technologies, such as hardware-based confidential computing or techniques such as differential privacy, to ensure that AI model building, validation, and analytics are conducted in a confidentiality- and integrity-preserving manner. The complexity of the specific technologies that enable this should be hidden from users, allowing them to focus on building AI models.

4.1.5. Robustness to heterogeneous computing systems. FL systems aim to enhance data availability from various sources. To do this, a federation should support low-resource environments that may lack access to costly infrastructure. This diversity can lead to several challenges that a federated system must manage. Training models in resource-limited settings can take much longer than in high-performance environments. If the system is configured not to wait for updates from slower sites, those sites might never contribute their data to model updates. Variations in computing infrastructure can also complicate the enforcement of security policies. For example, not all sites in a federation may have access to confidential computing or TEEs. Thus, enforcement of trusted computing consistently across the federation can be challenging.

In addition, FL systems must be designed to handle communication interruptions or outages. Algorithms should be resilient to situations where updates from some sites arrive late or not at all. This includes mechanisms to deal with stragglers, tolerate missing contributions, and resume operations smoothly after disruptions, ensuring robustness and progress despite unstable network conditions.

4.1.6. Increasing trust through enforced governance. Building trust among participants is crucial for scaling federations, which should be achieved through effective system governance. Different federations may have different security and privacy needs. Data custodians need assurance that their data will not be leaked through model parameter updates, and model developers need confidence that their IP will not be compromised at any data custodian site. All participants need to have confidence that all other participants are trustworthy. An FL system should enable users to set the security level appropriate to the threat model, which can then be enforced with privacy-enhancing technologies. For instance, an enterprise model developer might prioritize protecting their model's IP and enforce the use of TEEs, requiring system attestation (ensuring all compute nodes on a federation are legitimate) and workload attestation (verifying that all sites are running only the authorized workload) at each data custodian site. On the other hand, in contexts such as academic research, where the goal is to open-source the resulting model, protecting the IP of data custodians may be less critical, and enforcing TEEs may be optional.

4.1.7. Interoperability between federated learning frameworks. Several open-source FL frameworks are currently available for FL simulations and to implement FL in real-world and production environments, such as¹ FATE (91), Fed-BioMed (92), FederatedScope (93), FedML (94), Flower (95), FLUTE (96), HP Swarm Learning (97), IBM Federated Learning (98), NVIDIA FLARE (99), OpenFL (100, 101), PySyft (102), Substra (103), and TensorFlow Federated (104) to name a few in this growing field. Many federations start with any one of these many open-source frameworks. Often, these frameworks go through approvals at each data custodian's information technology and information security department. Interoperability between frameworks is essential to enable multiple federations on different frameworks to be combined, creating a federation of federations without placing an onus on federation participants by mandating them to replace their bespoke FL frameworks.

¹Noncomprehensive list in alphabetical order.

4.2. Business Considerations

FL involves multiple stakeholders who deliver and receive value. When developing a business model for FL, several factors must be considered to ensure sustainability. Value generation depends on data providers sharing sensitive data with model developers for collaborative AI training. In commercial or enterprise federations, decision-makers need cost-benefit and risk analyses to compare federated approaches with centralized ones. The business model should consider the following factors.

4.2.1. Data custodian's operational costs. Data providers incur expenses to extract, transform, and load data into a system accessible to the FL node. Additionally, they must be harmonized across data providers before training a model can begin. Setting up FL in privacy-sensitive industries requires a robust legal framework and, oftentimes, data-use agreements. Data custodians may be required to label data for supervised learning FL experiments. Onboarding and technical support for partner sites are necessary to maintain engagement. This burden often requires reimbursement or incentives for participants.

4.2.2. Infrastructure and computational costs. A federated system requires AI-capable computations at each data custodian's site. Depending on the use case, this can require expensive high-performance systems at every site. Additionally, centralized computing will be needed for aggregation, governance, and coordination. Implementing encryption, differential privacy, homomorphic encryption, or TEE-based solutions further increases computation costs. Although FL decreases raw data transfer, it still incurs bandwidth costs for model updates. FL-as-a-service (FLaaS) can often reduce costs by eliminating the need to build and maintain complex FL infrastructure from scratch. It offers built-in security and privacy compliance, reducing regulatory and legal overhead. FLaaS providers deliver on-demand computational infrastructure through cloud-native systems with optimized load balancing. They offer scalable pricing, so federations and underlying infrastructure can grow seamlessly and organizations pay only for compute cycles used in federated rounds.

4.2.3. Incentives for data participants. Model developers must create incentives for data custodians, such as reimbursement, usage-based pricing, or revenue sharing based on data contribution or model improvement. Furthermore, clear data terms of use and retention policies are crucial for the regulatory compliance and reproducibility of training experiments.

4.2.4. Model intellectual property. The business model must address governance issues such as who owns the models' IP and weights, and how updates are managed.

4.3. Scaling Federated Learning from Simulations to the Real World

Since its first application to healthcare, in 2018, as a simulated study for brain tumor segmentation (10), FL has been evaluated in numerous simulated and real-world studies. Example simulated use cases include brain anomaly detection (105), pancreas segmentation (106, 107), skin lesion classification (108), COVID-19 detection (109) and segmentation (110), and pan-cancer tumor infiltrating lymphocyte classification (111), to name a few.

Real-world FL studies have shared numerous insights and moved the field forward through small- and large-scale efforts. Examples of such small-scale, yet real-world, studies include training FL consensus models for (a) breast density classification based on mammography scans and according to the Breast Imaging, Reporting & Data System (BI-RADS) (112), (b) head computed tomography hemorrhage segmentation (113), and (c) whole prostate segmentation on MRI (114). Large-scale biomedical AI studies comprise (a) chest X-ray and EHR analysis for

COVID-19 patients (70), (b) the largest to-date real-world FL study involving 71 international sites that trained a global consensus model to detect boundaries of brain glioblastoma in MRI scans of >6,300 patients (63), and (c) using this federated neuro-oncology consortium to perform large-scale in-the-wild model validation (115).

These large-scale studies have disseminated practical considerations and valuable insights garnered during their global coordination efforts, contributing toward future implementations of more effective and efficient real-world FL. Key lessons from these studies include (a) the use of open-source infrastructure to increase the transparency, reproducibility, and extensibility of FL systems; (b) the value of partnerships between academia and industry, which bring together complementary expertise and resources; and (c) the importance of starting federations with trusted collaborators that share common goals and clear incentives for all involved parties at the outset. During early experimentation, all participants must plan and communicate frequently for feedback and troubleshooting. For successful model convergence, training data must be standardized and labeled consistently across all collaborators before training begins. In addition, infrastructure requirements should be clearly articulated, including a minimum and recommended configuration for compute infrastructure and network requirements.

To scale FL beyond the scope of those studies and pave the way for its consistent usage in practice, we need to overcome challenges on both the technical and business sides. Technically, FL demands a robust infrastructure to handle system heterogeneity, mitigate adversarial attacks to increase trust, and ensure ease of system use for all persons to lower the barriers for data custodians to join a federation. On the business side, a federation must align incentives between participants, ensure data privacy compliance across jurisdictions, and justify return on investment. Addressing these challenges requires cross-disciplinary collaboration, thoughtful platform design, and a sustainable business model that balances value creation with privacy and trust. FL systems are ultimately expected to deliver return on investment by enabling new use cases that overcome data silos and by enhancing model utility.

4.4. Future Research Directions

Although substantial advances have been made in FL research over the past few years (see Sections 2 and 3), critical challenges persist that hinder its large-scale deployment and clinical translation. Addressing the following targeted areas will be key to bridging this gap:

- Learning under non-IID conditions: Statistical heterogeneity remains one of the most fundamental obstacles in biomedical FL. There is a need for systematic study of personalized aggregation, meta-learning, and domain adaptation strategies that allow models to balance global coherence with local specificity, particularly when site-level distributions differ in patient demographics, imaging protocols, or disease prevalence.
- Quantification and mitigation of data heterogeneity: Open questions remain on how to measure the degree of non-IIDness across clients and how such measures correlate with convergence stability, bias, and generalization. Developing standardized metrics and adaptive regularization approaches could provide deeper insights into model behavior across distributed datasets.
- Integrating FL with large models: In recent years, large foundation models have shown remarkable success in diverse biomedical applications, from clinical (radiology and pathology) report generation to pathology image interpretation and EHR summarization. However, their training often relies on centralized data sharing, which is impractical or unethical for sensitive clinical data. A promising future direction lies in integrating FL with these large models to enable collaborative training and inference across institutions without data

sharing. This integration will require innovations in efficient federated fine-tuning to reduce communication and computational costs.

- **Fairness and bias-aware aggregation:** Ensuring fairness in FL requires aggregation methods that mitigate biases arising from uneven data quality, client participation, and temporal drift. Incorporating measures of uncertainty, data representativeness, and client reliability into the aggregation process can help balance model influence across participants, promoting equitable and trustworthy learning outcomes.
- **Federated validation and benchmarking:** Systematic evaluation protocols that preserve data privacy while enabling cross-site validation are still not widely considered. Designing federated evaluation frameworks (90) with harmonized metrics and privacy-preserving validation mechanisms will be crucial for assessing model performance and reproducibility in real-world settings.
- **Privacy preservation and secure computation:** While FL reduces the need for data sharing, it remains vulnerable to information leakage through gradients, updates, or metadata. Future research should explore the combination of differential privacy, secure multiparty computation, and homomorphic encryption techniques with hardware-based confidential computing and TEEs that offer rigorous privacy guarantees while maintaining model utility. Balancing computational overhead, privacy budgets, scalability, and model utility remains a pressing open question.
- **Operational and governance frameworks:** Persistent FL infrastructures must incorporate life-cycle management, data provenance, security guarantees, and compliance monitoring to ensure trustworthy deployment in regulated healthcare environments. Research on standardized interfaces and reproducibility protocols can facilitate interoperability and long-term sustainability.

Addressing these challenges will accelerate the transition from experimental FL studies to clinically actionable systems. By combining algorithmic innovation with robust infrastructure and governance, FL can enable the development of equitable, generalizable, and continuously improving AI models that deliver measurable clinical benefit.

5. CONCLUSIONS

In conclusion, it has become apparent that while training computational/AI models for biomedical engineering, there is a need to (a) address the data heterogeneity issues across clients; (b) assess algorithmic generalizability; and (c) address automatic quality control, monitoring of drift and bias, and model calibration. FL offers a paradigm shift in multi-institutional collaborations that (a) enables and even expedites access to unprecedented amounts of data from diverse patient populations; (b) offers a solution to legal, privacy, and ownership concerns that currently hamper collaborations based on CL; (c) performs on a par with CL model training; and (d) can facilitate postclearance monitoring of model utility over time. Major algorithmic advances have been observed and are still actively being developed that relate to improving the efficiency, functionality, and fairness of FL. Security and privacy should be prime considerations to ensure that FL-related assets are always protected. Despite the progress we have seen in the field of FL, moving from simulated to large-scale real-world studies (63, 70, 115), there is a need for complete services (FLaaS) to move FL from one-off investigator-initiated studies toward persistent FL for AI studies in healthcare. Achieving this can yield a paradigm for robust generalizable AI models, with accelerated discovery and real clinical impact, from drug discovery to clinical diagnostic, prognostic, and predictive healthcare AI models and medical devices.

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